Report for Viz Lab 2a

Overview
In today's world of trends and social interactions, individuals rely on their daily infusion of content from various online platforms. Working with this dataset featuring trendy videos was enjoyable. The amalgamation of these attributes guarantees a comprehensive exploration of YouTube channels, covering aspects such as popularity, financial dynamics, and growth patterns. This contributes to a holistic analysis of the digital content landscape in 2023.

The assignment is based on the youtube statistics of 2023 globally on the ranks of the videos and the views of the same, it depends mainly on the popularity factor, the dataset i have chosen here is the "Global YouTube Statistics 2023", this dataset gives a snapshot of YouTube channels worldwide. It covers essential details like rankings, subscribers, video views, etc. This data helps understand where channels are popular, how much money they might earn, and how they've evolved over time. It's a valuable resource for insights into the global YouTube landscape.

I plan to utilize the dataset from the previous assignment, focusing solely on numerical data, and carry out all the necessary tasks based on this dataset.

Dataset
Name of the dataset - Global YouTube Statistics 2023
Source of the dataset - Kaggle.com
URL – https://www.kaggle.com/datasets/nelgiriyewithana/global-youtube-statistics-2023

Number of attributes in the dataset initially - 16
Number of attributes in the dataset considered for this assignment - 8
Number of data points - 588

Attributes (8):

- **subscribers**: Number of subscribers to the channel
- **video views**: Total views across all videos on the channel
- **uploads**: Total number of videos uploaded on the channel
- **video_views_for_the_last_30_days**: Total video views in the last 30 days
- **lowest_yearly_earnings**: Lowest estimated yearly earnings from the channel
- **highest_yearly_earnings**: Highest estimated yearly earnings from the channel
- **subscribers_for_last_30_days**: Number of new subscribers gained in the last 30 days

- **created_year**: Year when the YouTube channel was created

Implementation

The first two principal components derived through PCA efficiently capture over 60% of the underlying patterns in the data. This insight underscores the practicality of representing complex datasets, potentially laden with numerous dimensions, by condensing them into just two principal components. Principal Component Analysis (PCA) proves particularly beneficial in simplifying data, showcasing its utility in achieving a substantial intrinsic dimensionality index of 75% accuracy through the utilization of only two to three principal components. This emphasizes the method's effectiveness in clarifying essential information and reducing the complexity of datasets.

The mechanics of PCA's "fit transform" feature involve computing the dot product between an N-dimensional data point and each PCA component when the data is scaled. This operation contributes to the effective transformation and representation of the data in a more manageable form. Furthermore, the visual representation of the PCA biplot sheds light on the most influential dimensions along the first two principal components, with a clear emphasis on User Score, Critic Score, and Global Sales. This graphical insight aids in the interpretation of key factors steering the data's variability.

Analyzing the elbow point in a graph, specifically at k=2, where the curve exhibits a noticeable flattening and the mean squared error distortion is low, guides the informed choice of k=2 for subsequent k-means clustering. Additionally, the scatter plot matrix, influenced by the chosen intrinsic dimensionality index, provides a concise visual summary of the top four attributes, focusing on their squared loadings.

When putting the plan into action, using Python for the backend API development adds a hands-on element to the project. This not only allows for the exploration of Flask but also involves addressing and navigating a few Cross-Origin Resource Sharing (CORS) challenges, enriching the learning experience. Incorporating a code base from the previous assignment, on building scatter plots, PCA biplots, and scatter matrices, serves as a foundational framework. Despite encountering minor issues, the endeavor to seamlessly integrate the backend API with d3.js proves to be both intriguing and instructive, offering valuable insights into interactive data visualization.

Conclusion

The exploration of the 'Global YouTube Statistics 2023' dataset has yielded profound insights into the dynamics of global YouTube channels. Through interactive plots and user-friendly features, the web page facilitates a comprehensive exploration of diverse numerical attributes. The inclusion of a dynamic navigation button enhances user experience, allowing seamless transitions between pages and customization of plot views. This dataset emerges as a valuable asset for stakeholders, equipping them with actionable insights and a strategic foundation to navigate the dynamic realm of online content creation.